

United States Department of Agriculture
Project Initiation

Project Data for Accession Number 1034425

Project Title

DSFAS: AI-driven genomic and crop simulation models for enhancing plant breeding using high-throughput phenotyping

Project Details

Sponsoring Institution National Institute of Food and Agriculture	Project Status ACTIVE
Funding Source NIFA Non Formula	Grant Year 2026
Grant No. 2026-67021-45864	Cumulative Award Amt. \$591,460.00
Proposal No. 2024-13400	Multistate No. (N/A)
Project Start Date Feb 01, 2026	Project End Date Jan 31, 2029
Program Code [A1541] Food and Agriculture Cyberinformatics and Tools	

Project Director

Zotarelli, L.

Recipient Organization

UNIVERSITY OF FLORIDA
1523 UNION RD RM 207
GAINESVILLE, FLORIDA 326111941

Performing Department

(N/A)

Non Technical Summary

The project aims to speed up the development of potato varieties that can thrive in the hot, humid climate of the southern U.S. Traditional breeding cycles can take ten years or more, but the team plans to cut that time by combining three modern tools: artificial intelligence (AI), high-throughput phenotyping (HTP), and crop-simulation models linked to whole-genome prediction (WGP). First, an inexpensive RGB video camera will be installed on the post-harvest grading line. The camera captures every tuber as it moves through the sorter, and AI models will automatically measure size, weight, skin color and detect defects such as cracks or greening. Second, unmanned aerial vehicles (UAVs) and high-resolution satellite imagery will be used to collect multispectral, hyperspectral and 3D point-cloud data from the field. AI algorithms that fuse these multiple sensor streams will estimate above-ground biomass, tuber biomass, nitrogen uptake and yield throughout the growing season. Third, the AI-derived trait estimates will be fed

United States Department of Agriculture
Project Initiation

into a potato crop?simulation model (e.g., DSSAT?SUBSTOR?potato model) that has been enhanced to handle heat?stress and specific?gravity effects typical of subtropical conditions. Further calibration of genotype?specific parameters, and the calibrated model will be combined with whole?genome prediction to forecast how each genotype will perform across a range of environments, management practices and climate scenarios. The work will be carried out over three years on more than 7,000 experimental plots representing over 2,000 distinct genotypes. Data collection, model development and validation are organized into a series of experiments that progress from video?based tuber phenotyping, through UAV?based field phenotyping, to the final integrated AI?CSM?WGP framework. All code, models and curated datasets will be released openly on repositories, and the team will publish findings in open?access journals and present them at conferences. By providing rapid, low?cost, and highly accurate phenotypic information, the project is expected to (1) shorten the breeding cycle, (2) lower labor and research expenses, (3) improve the physiological realism of crop models under climate?stress conditions, and (4) deliver new, high?yielding, heat?tolerant potato varieties that support the long?term sustainability of the potato industry and contribute to national food?security goals. The approach also creates a training ground for post?doctoral researchers and graduate students in AI, remote sensing, crop modeling and genomics, building a skilled workforce that can apply these methods to other crops and regions.

Animal Health Component

Animal Health Component: 0%

Research Effort Categories

Basic	20%
Applied	50%
Developmental	30%

Classification

Knowledge Area (KA)	Subject of Investigation (SOI)	Field of Science (FOS)	Percent
201	1310	1081	50%
201	1310	2080	50%

Knowledge Area

[201] Plant Genome, Genetics, and Genetic Mechanisms

Subject Of Investigation

[1310] Potato

Field Of Science

[1081] Breeding

[2080] Mathematics and computer sciences

Keywords

AI; Automation; Breeding; Crop Models, High throughput phenotyping; Machine learning; Potato

United States Department of Agriculture Project Initiation

Goals / Objectives

The proposal seeks to accelerate the development of potato varieties for subtropical regions by tightly integrating artificial intelligence (AI) methods, high-throughput phenotyping (HTP), crop simulation models (CSM), and whole genome prediction (WGP). Its first objective is to create advanced AI algorithms that can perform real time, postharvest phenotyping on the grading line, capturing traits such as tuber size, weight, skin color, and deformities while reducing the current labor intensive workflow. The second objective focuses on developing AI driven, in field phenotyping pipelines that fuse multispectral (MSP), hyperspectral (HSP), three dimensional point cloud, and high resolution satellite data collected with UAVs; these multi modal architectures will estimate above ground biomass, tuber biomass, nutrient uptake, and stress indicators for thousands of genotypes across diverse environments. The third objective is to integrate the AI derived traits with CSMs and WGP to model genotype $\times$ environment $\times$ management (G $\times$ E $\times$ M) interactions, calibrating genotype specific parameters and refining yield and quality predictions under heat stress and humidity conditions typical of the southern United States. Together, these efforts aim to shorten the traditional ten year breeding cycle, lower phenotyping costs and labor, improve the physiological fidelity of crop models in a changing climate, and deliver high yield, heat tolerant potato cultivars that support the long term sustainability of the national potato industry.

Project Methods

The project will be carried out over three years using an integrated workflow that combines high-throughput phenotyping (HTP), advanced artificial intelligence (AI) algorithms, crop simulation modeling (CSM), and whole genome prediction (WGP). Data acquisition in the post-harvest grading line using a low-cost RGB video camera will be installed to capture every tuber as it moves through the sorting system. The video is collected each harvest season for all ~7,000 experimental plots, and ground-truth measurements (tuber length, width, weight, eye depth, skin color, and deformities) are recorded by trained staff to create a calibrated training set. In the field, unmanned aerial vehicles (UAVs) will fly at key growth stages to acquire multispectral (MSP), hyperspectral (HSP) and high-resolution 3D point-cloud data; complementary satellite MSP imagery (~50cm resolution) will be downloaded for each trial. Ground-truth biomass, tuber yield, specific gravity, nutrient status and agronomic management records are taken simultaneously, providing a multimodal dataset spanning genotype $\times$ environment $\times$ management (G $\times$ E $\times$ M) conditions. The AI modeling workflow is organized into three sequential steps. First, a hierarchical classifier is built to separate "deformed" from "non-deformed" tubers and then to identify specific defect types; the system is trained on grading-line video and a contrastive semantic-segmentation network, with generative adversarial augmentation used to balance the data and achieve pixel-level defect localization. Second, multimodal temporal fusion or temporal graph neural networks ingest UAV-collected and satellite data; transformer-based or temporal convolutional architectures capture cross-sensor interactions and phenological dynamics, producing estimates of above-ground biomass, tuber biomass, nitrogen uptake, and yield. Third, the AI-derived trait estimates are integrated into the DSSAT-SUBSTOR potato simulation (or, when appropriate, an asymptotic growth model) that has been expanded with heat-stress functions and specific gravity parameters; Approximate Bayesian Computation calibrates genotype-specific coefficients for the model by minimizing the root-mean-square error between simulated and observed yields. The calibrated CSM outputs are then combined with genomic marker data in a whole-genome prediction framework. Mixed linear models implement WGP, while the AI-enhanced CSM supplies additional proxy phenotypes, improving prediction accuracy for untested environments. Experimental validation follows a multi-environment trial design: each year the breeding program will evaluate ~2,000 distinct genotypes across the major subtropical production zones. Model performance is quantified by coefficient of determination (R^2), root-mean-square error (RMSE) and classification metrics (precision, recall) for deformity detection, and by genetic gain per cycle for the integrated CSM-WGP pipeline.